MIL-W-3318A
22 February 1971

SUPERSEDING
MIL-W-3318
25 October 1950

MILITARY SPECIFICATION

WIRE, COPPER; AND WIRE, STEEL, COPPERCLAD
(W-154 AND WS-24/U)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers tie wires, 19 inches long, 0.080 inches diameter, which are used for attaching copper or copperclad steel line wires to insulators. The copper tie wire is designated as wire W-154 and the copperclad steel tie wire is designated as wire WS-24/U.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

FEDERAL

SPECIFICATIONS

QQ-S-781	Steel Strapping, Flat.
PPP-B-585	Boxes, Wood, Wirewound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grades) and Cut Shapes.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

FSC 9505

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

MILITARY

SPECIFICATIONS

MIL-P-116	Preservation, Methods Of.

STANDARDS

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification, to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee (UCC)

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

National Motor Freight Traffic Association, Inc. (NMFTA)

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, 1616 P Street, N.W., Washington, D.C. 20036.)

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3. REQUIREMENTS

3.1 Material. The material of the tie wires shall be as specified herein. The best material for tropical and arctic use commercially available for the purpose shall be used when a definite material is not designated.

3.1.1 Copper tie wire. The material used in the manufacture of this wire shall be copper of such quality and purity that when drawn and annealed the finished wire shall conform to this specification.

3.1.2 Copperclad steel tie wire. The material used in the manufacture of this wire shall be soft copperclad steel wire of the grade designated commercially as "40 percent conductivity". The wire shall be drawn and annealed in such a manner that the finished tie wire shall conform to this specification.

3.2 Physical requirements. All tie wires shall be bright annealed. They shall show no evidence of burning or other injury received in the annealing process. When the annealing is done after the tie wires have been assembled in bundles, it shall be so controlled as to insure that all wires throughout the bundles are uniformly annealed and that they conform to this specification.

3.2.1 Diameter and length. All tie wires shall have a diameter of 0.080 $\pm$0.004 inch and an overall length of 19 inches $\pm$0.25 inch. (See 4.3.)

3.2.2 Tensile strength, breaking strength, and elongation. Tensile strength, breaking strength and elongation shall be as specified in table I when tested as specified in 4.4. The maximum allowable breaking strength for wire differing from the nominal size, but within the tolerances specified, shall be determined by multiplying the applicable value of tensile strength in pound-force per square inch (lb_f/in^2), as specified in table I, by the cross-sectional area (square inch) of the sample.

Table I. Tensile strength, breaking strength and elongation.

Material	Nominal diameter	Tensile strength (maximum)	Breaking strength (maximum)	Elongation (minimum)
	Inch	Lb_f/in^2	Pounds	Percent in 10 inches
Copper - - - - - -	0.080	39,500	199	25
Copperclad steel -	0.080	70,000	352	15

3.2.2.1 Elongation. The elongation shall be determined as the permanent increase in length, at the time of failure of the wires in tension, measured between bench marks originally placed on the wire 10 inches apart. Elongation determinations shall be disregarded in cases where the break is not located between the bench marks and at least 1 inch from either bench mark.

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3.2.3 Ductility of copperclad steel and copper wire. (See 4.5.)

3.2.3.1 Torsion. The finished wire shall not fracture when tested as specified in 4.5.1.

3.2.3.2 Bending. The finished wire shall not fracture when tested as specified in 4.5.2.

3.2.4 Bonding of copper to steel. For copperclad steel wire, the copper shall not separate from the steel when tested as specified in 4.6.

3.3 Workmanship. The wire shall be manufactured and finished in a thoroughly workmanlike manner in accordance with the best commercial practice of wire manufacturers. All finished tie wires shall be cylindrical in form and free from scales, inequalities, splints, or other flaws. Small surface imperfections, such as longitudinal die marks, scratches, and minute fissures resulting from removal of small splints, are permissible provided that such imperfections do not exceed those characteristics of good manufacturing practice. The ends of the wires shall be smoothly cut so as to be free of sharp, chisel-shaped, or ragged ends.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Selection of samples. Three sample bundles shall be taken at random from each lot of 100 bundles, or fraction thereof, (but not less than five sample bundles from the complete order.) Each bundle represents 10 pounds of tie wire. From each bundle thus selected, one sample tie wire shall be taken for each of the following tests.

4.3 Diameter and length. Samples selected in accordance with 4.2 shall be inspected for compliance with the diameter and length requirements of 3.2.1.

4.4 Tensile strength, breaking strength, and elongation. Samples selected in accordance with 4.2 shall be tested on a standard materials testing machine for compliance with 3.2.2.

4.5 Ductility tests. (See 3.2.3.) Samples selected in accordance with 4.2 shall be tested as follows:

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4.5.1 Torsion. Each sample shall be subjected to 25 twists in a length equivalent to 100 times the diameter of the wire under test. This test shall be made in a standard torsion machine or equivalent, with one head of the machine rotating and the opposite head nonrotating but capable of moving horizontally. The weight on the movable head shall be 15 pounds, and the rate of rotation approximately 15 twists per minute. After completion of the testing, the samples shall meet the requirements of 3.2.3.1.

4.5.2 Bending. Each sample shall be subjected to 15 bends through an angle of 180 degrees over a 3/16-inch radius. A bend shall be defined as the movement of the wire through an angle of 90 degrees in either direction from the original position. The bends shall be made at a uniform rate of not more than 60 bends per minute. At the completion of the testing, the samples shall meet the requirements of 3.2.3.2.

4.6 Bonding test on copperclad steel wire. At the conclusion of the torsion and bending tests specified in 4.5.1 and 4.5.2, each sample of copperclad steel wire used in these tests shall be tested further until the wire fractures. The fracture shall be examined visually to insure that the sample conforms to 3.2.4.

4.7 Lot acceptance criteria. The lot shall be accepted if no failures occur in the sample.

4.8 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Unless otherwise specified, preservation and packaging shall be level A or C, as specified. (See 6.1(c).)

5.1.1 Level A.

5.1.1.1 Cleaning. Tie wires shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Tie wires shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Packaging. The tie wires shall be packaged method III of MIL-P-116. Tie wires shall be prepared in bundles of 10 pounds each, the bundles shall be secured with soft copper or copperclad steel tie wires. Place the bundled tie wires individually within a close-fitting container conforming to PPP-B-636, grade W5c. Close the box in accordance with the appendix of the box specification.

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5.1.2 Level C. Tie wires shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage. The supplier's standard practice is acceptable when it fulfills these requirements.

5.2 Packing. Packing shall be level A or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.1(c).)

5.2.1 Level A.

5.2.1.1 Consolidation. A quantity of tie wires, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix to the box specification. To facilitate palletization, fiberboard boxes shall be unform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. A quantity of containers, packed as specified in 5.2.1.1 shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.

5.2.2.1 Consolidation. A quantity of tie wires packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load. A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

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5.2.2.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C.

5.2.3.1 Consolidation. A quantity of tie wires packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

5.2.3.2 Palletized load. A quantity of containers, packed as specified in 5.2.3.1 shall be palletized as specified in 5.2.1.2, except that fiberboard caps shall be class domestic.

5.2.3.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.2.3.4 Alternate method of packing. The wire shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:
(a) Title, number and date of this specification.
(b) Type required. (See 1.1, 3.1.1, and 3.1.2.)
(c) Levels of preservation, packaging, packing and marking. (See Section 5.)
(d) When the rough handling and functional tests are required.

6.2 Indirect shipments. The preservation, packaging, packing and marking specified in section 5 apply only to direct purchases by or direct shipments to the government, and are not intended to apply to contracts or orders between the supplier and prime contractor.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-W-3318A, Wire, Copper; and Wire,Steel,Copperclad (W-154 and WS-24/U)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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2.6 Workmanship. The finished dispensers shall be complete, clean, and free

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